INCH-POUND

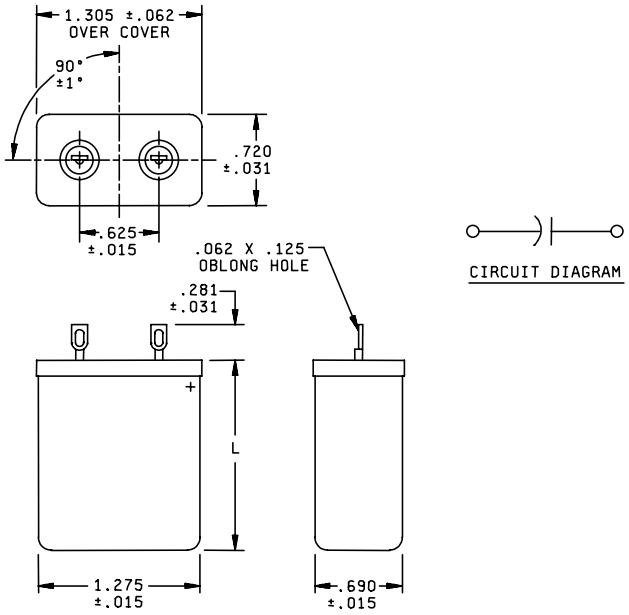
MIL-DTL-3965/21F
19 August 2002
SUPERSEDING
MIL-C-3965/21E
8 June 1972

DETAIL SPECIFICATION

CAPACITOR, FIXED ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG) 85°C (VOLTAGE DERATED TO 125°C) STYLE CL55 GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW EQUIPMENT DESIGN.
NO SUPERSEDING DOCUMENT.



	Dimensions	
	L ±.031 (.79)	
	In	mm
A1	1.062	26.97
A2	1.375	34.93
A3	1.625	41.28
A4	2.000	50.80
A5	2.500	63.50

Inches	mm
.015	.38
.031	.79
.062	1.57
.125	3.18
.281	7.14
.625	15.88
.690	17.53
.720	18.29
1.275	32.39
1.305	33.15

NOTES:
1. All dimensions in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style CL55 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Capacitor characteristics.

PIN	Voltage			Capaci-tance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL55BE961MPG	15	10	17.2	960	7	58	3.10	A1
CL55BE122MPG	15	10	17.2	1200	9	72	2.30	A2
CL55BE142MPG	15	10	17.2	1400	11	84	1.70	A3
CL55BE212MPG	15	10	17.2	2100	16	126	1.30	A4
CL55BE242MPG	15	10	17.2	2400	18	144	1.15	A5
CL55BH521MPG	30	20	34.5	520	8	63	5.30	A1
CL55BH661MPG	30	20	34.5	660	10	80	4.20	A2
CL55BH821MPG	30	20	34.5	820	13	99	2.90	A3
CL55BH122MPG	30	20	34.5	1200	18	144	2.30	A4
CL55BH132MPG	30	20	34.5	1300	20	156	2.10	A5
CL55BJ401MPG	50	30	57.5	400	10	80	7.20	A1
CL55BJ431MPG	50	30	57.5	430	10	78	6.40	A1
CL55BJ501MPG	50	30	57.5	500	13	100	5.60	A2
CL55BJ601MPG	50	30	57.5	600	15	120	4.10	A3
CL55BJ801MPG	50	30	57.5	800	20	160	3.10	A4
CL55BJ102MPG	50	30	57.5	1000	23	180	2.80	A5
CL55BL271MPG	75	50	86.2	270	9	81	8.50	A1
CL55BL331MPG	75	50	86.2	330	12	99	7.00	A2
CL55BL401MPG	75	50	86.2	400	15	119	5.00	A3
CL55BL601MPG	75	50	86.2	600	23	180	3.65	A4
CL55BL661MPG	75	50	86.2	660	25	198	3.50	A5
CL55BN171MPG	100	65	115.0	170	9	68	15.00	A1
CL55BN221MPG	100	65	115.0	220	11	88	11.60	A2
CL55BN261MPG	100	65	115.0	260	13	104	8.00	A3
CL55BN351MPG	100	65	115.0	350	18	140	6.50	A4
CL55BN441MPG	100	65	115.0	440	22	176	5.80	A5
CL55BQ700MPG	150	100	172.0	70	6	42	28.80	A1
CL55BQ900MPG	150	100	172.0	90	7	54	22.40	A2
CL55BQ101MPG	150	100	172.0	100	8	60	16.40	A3
CL55BQ141MPG	150	100	172.0	140	11	84	12.40	A4
CL55BQ181MPG	150	100	172.0	180	14	104	11.20	A5

Periodic inspection: Group C inspection shall be in accordance with MIL-DTL-3965 except as follows:
Inspection schedule for each subgroup shall be every 12 months.

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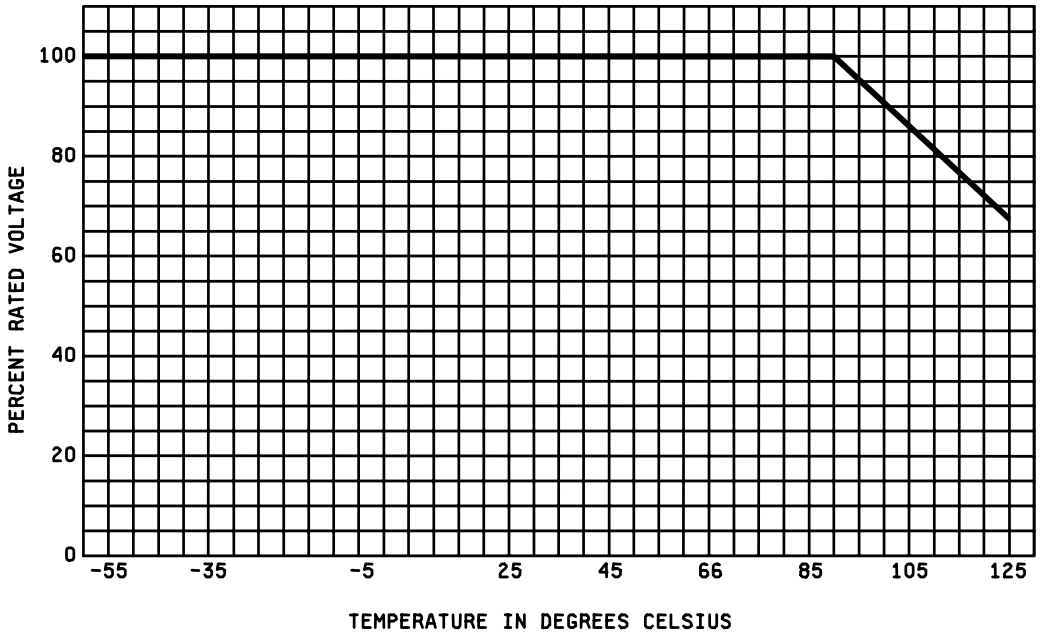

FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration:

Capacitors - See figure 1.

Brackets - See figure 3.

Case type - Rectangular, uninsulated

Seal type - Hermetic (symbol G).

Terminals - See figure 1.

Rated temperature - -55°C to +85°C,
voltage derated to +125°C (see figure 2).

Rated voltage - See table I.

Capacitance (cap) (nom) - See table I.

Capacitance tolerance - Symbol M (±20%).

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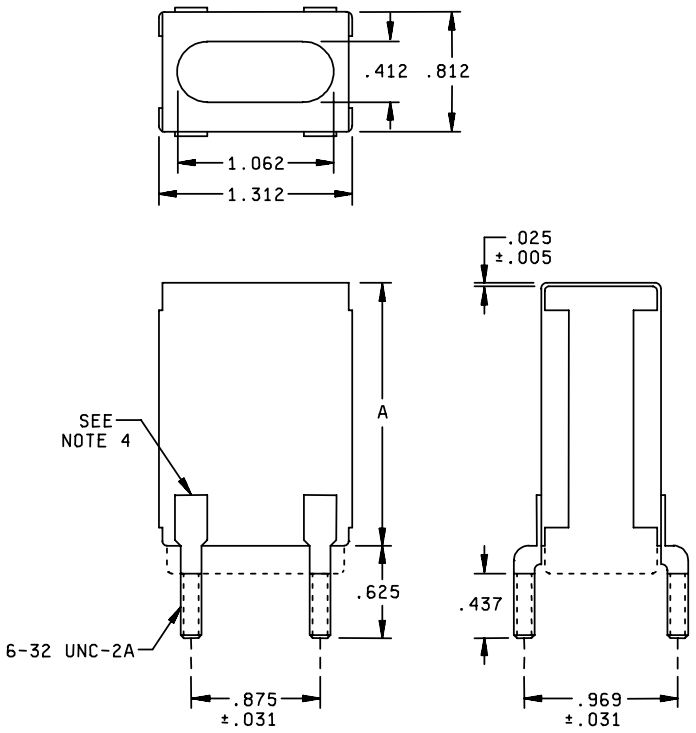
Inches	mm
.025	0.64
.412	10.46
.437	11.10
.625	15.88
.812	20.62
.875	22.23
.969	24.61
1.062	26.97
1.312	33.32



BRACKET FOR TYPE A CASE

Type Designation	Case Size	Dim	Dimensions				
			Inches			Millimeters	
			Min	Nom	Max	Min	Max
CL050SA13	A1	A	.846	.862	.878	21.49	22.30
	A2 1/	A	1.159	1.175	1.191	29.44	30.25
CL050SA33	A3	A	1.409	1.425	1.441	35.79	36.60
	A4 1/	A	1.784	1.800	1.816	45.31	46.13
	A5 1/	A	2.284	2.300	2.316	58.01	58.82

1/ CL050SD13, CL050SD23 and CL050SD33 brackets (of MIL-DTL-3965/9) may be used for case sizes A2, A4 and A5, respectively.

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.157 mm).

FIGURE 3. CL050 brackets.

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DC leakage (DCL) (max) (at +25°C):
See table I.

Dissipation factor (DF) (max):
15% (≤50 v) or 12% (>50 v).

Seal: In accordance with MIL-DTL-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 300 volts.

Insulation resistance:
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 100 megohms.

Temperature cycling:
DCL (max) - 125% of 25°C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: In accordance with method 208 of MIL-STD-202.

Terminal strength:
Pull test - In accordance with MIL-DTL-3965.
Bend test (if req.) - In accordance with MIL-DTL-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25°C value of table I.
ΔCap - Within ±8% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25°C)
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25°C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - See table I.

Step 2 (-55°C):
Impedance - See table I.
ΔCap (max) - -35% of step 1 value.

Step 3 (+25°C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.

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Step 4 (+85°C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.

Step 5 (+125°C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.

Step 6 (+25°C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage:

Life at +85°C:
2,000-hr (qualification):
DCL (max) - See table I (25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

240-hr (group B):
DCL (max) - See table I (25°C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.

1,760-hr (group C):
DCL (max) - See table I (25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

Life at +125°C:
DCL (max) - See table I (25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Marking: Capacitors shall be marked as shown in the following example:
Example:
CL55BE961MPG-- Type designation.
12345 7015 - source and date code.
960 µF 15 v dc - Capacitance and rated voltage (at 85°C).

Polarity - Plug (+) symbol near positive terminal (see figure 1).

Voltage group (85°C rated voltage):
I - - - - - - - - - - - - - 15 to 150 v, incl.

Substitution data. Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

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